

SOC Incident Response Case Study – Multi-Stage Attack Investigation

Case Information

- Date: 2026-01-02
- Analyst: Ibraim Marinos Lian
- Environment: Splunk (Ubuntu) + Windows 10 VM
- Objective: Simulate a multi-stage attack, correlate logs across sources, build a minute-by-minute timeline, and map to MITRE ATT&CK.

Incident Overview

While reviewing authentication activity in Splunk, I noticed an unusual increase in failed login attempts targeting a Windows endpoint.

At first, the activity could have been explained by a user entering an incorrect password. However, the volume and frequency of the failed authentication attempts suggested a possible brute-force attack.

To determine whether the activity progressed beyond failed logins, I expanded the investigation by reviewing Windows authentication events, PowerShell activity, and endpoint telemetry.

As the investigation continued, I identified additional indicators that suggested the attack had moved beyond credential access and into execution activity.

Initial Observations

During the early stages of the investigation, I identified several suspicious indicators:

- Multiple failed authentication attempts within a short time period
- A successful login following repeated failures
- Suspicious PowerShell activity
- Evidence of file download behavior
- Indicators of outbound communication

These observations suggested that the activity was likely part of a multi-stage attack rather than an isolated authentication issue.

Investigation Phase 1 – Authentication Analysis

The first step was to determine whether the failed login activity represented normal user behavior or an attempt to gain unauthorized access.

I reviewed authentication events in Splunk and focused on login failures occurring within a short timeframe.

The results revealed multiple failed login attempts against the same endpoint, supporting the hypothesis that credential access activity may have occurred.

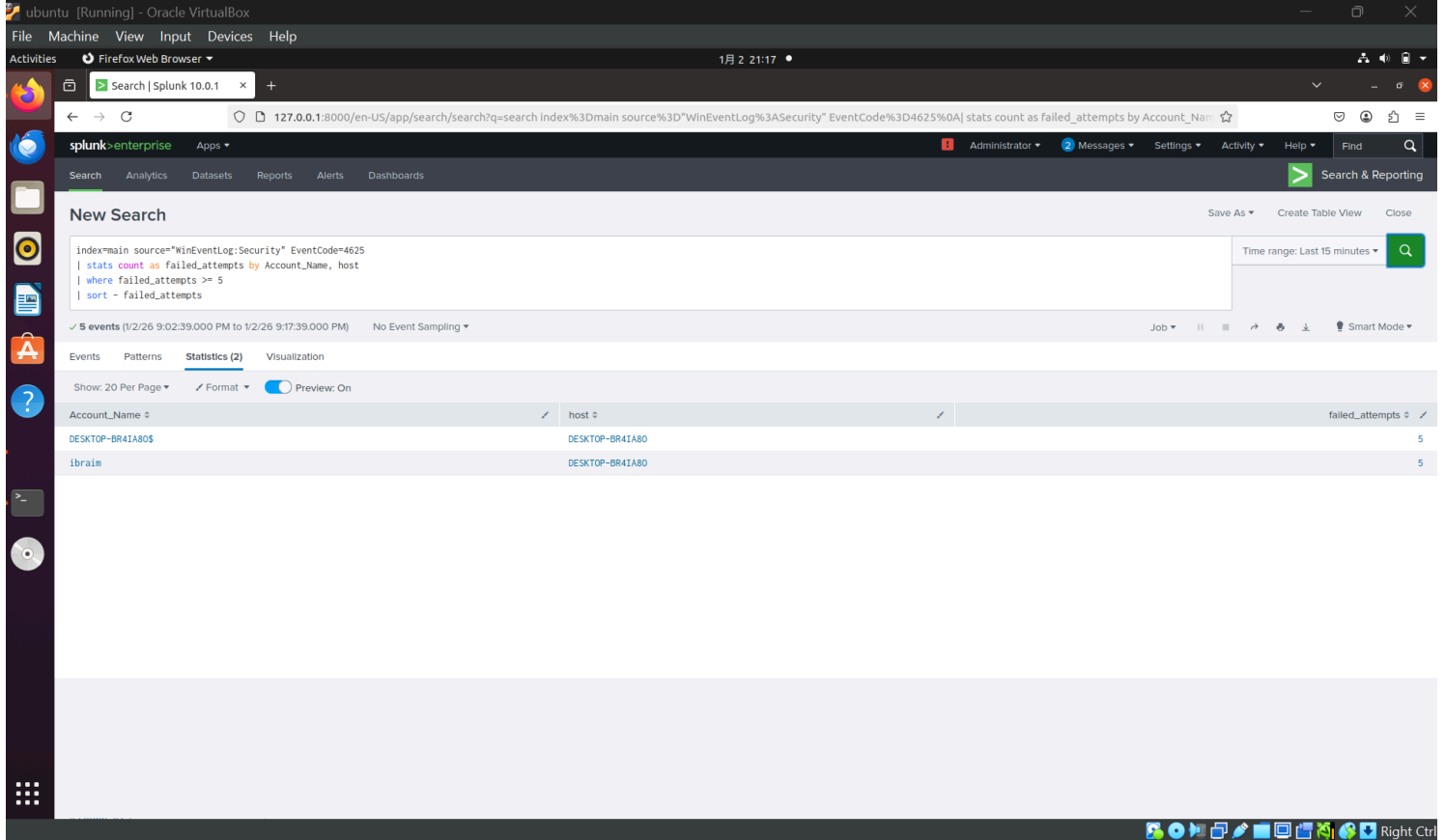


Figure 1: Multiple failed authentication attempts identified during the initial investigation phase, indicating potential credential access activity.

Investigation Phase 2 – Authentication Validation

After identifying multiple failed authentication attempts, I wanted to determine whether any account eventually authenticated successfully.

A successful login following repeated failures could indicate that valid credentials had been obtained or that an attacker successfully guessed a password.

I reviewed Windows Security Event ID 4624 events in Splunk and correlated them with the previously observed failed logon activity.

The results showed successful authentication events occurring on the same endpoint shortly after the failed login activity.

This finding increased confidence that the activity was not simply a user entering an incorrect password, but potentially part of a credential access sequence.

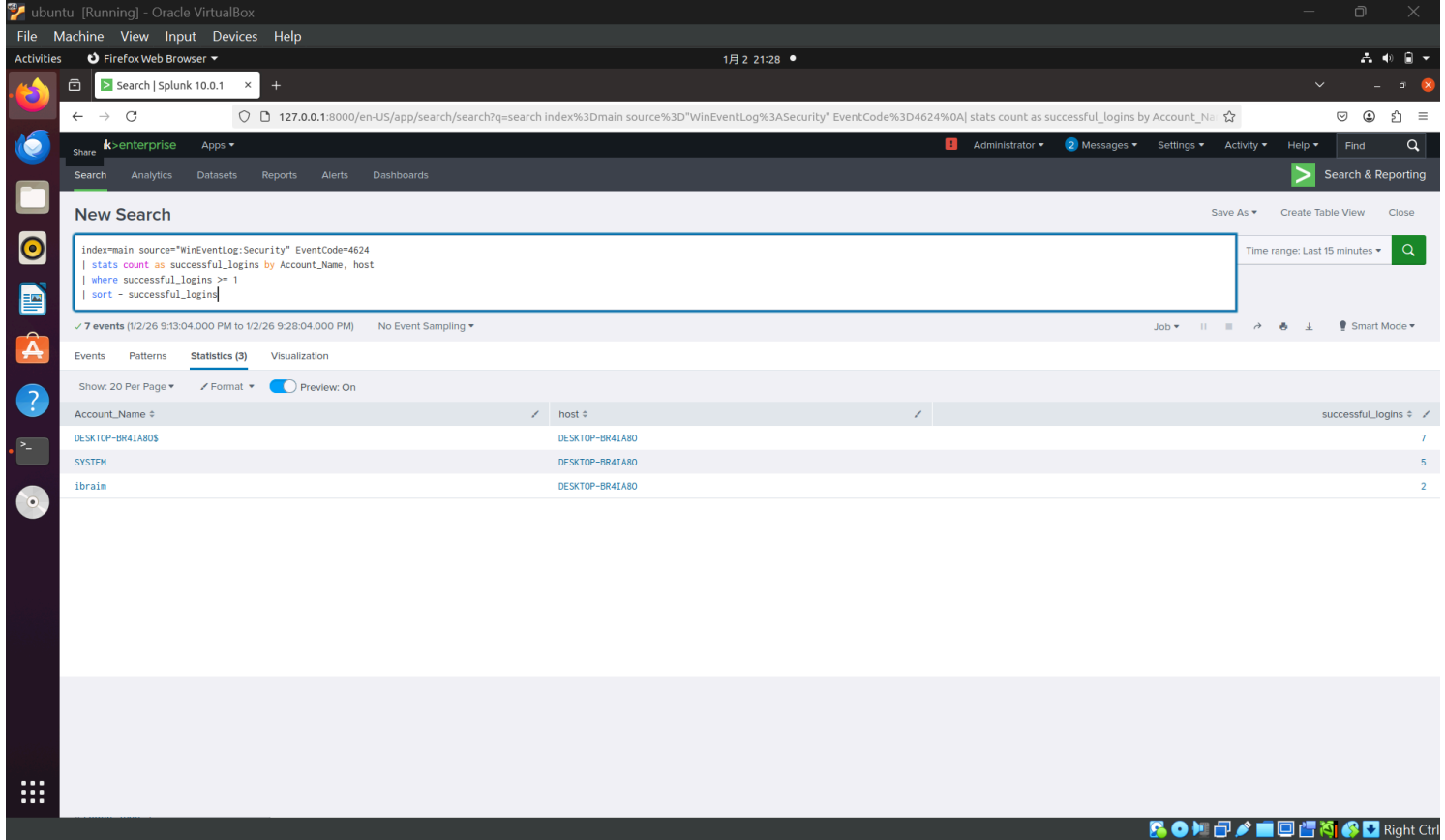


Figure 2: Successful authentication events identified following repeated login failures, suggesting possible credential compromise or successful credential discovery.

Investigation Phase 3 – PowerShell Activity Analysis

After confirming successful authentication activity, I wanted to determine whether any suspicious commands were executed on the endpoint.

A common next step after gaining access is the execution of PowerShell commands to download tools, establish persistence, or perform additional actions on the compromised system.

I reviewed PowerShell Operational logs (Event ID 4104) in Splunk and searched for indicators of script execution and file download activity.

The investigation revealed the use of Invoke-WebRequest, a PowerShell command commonly used to retrieve files from external sources.

The command attempted to download content and save it locally on the endpoint, suggesting post-compromise activity and possible tool deployment.

This finding indicated that the attack had progressed beyond credential access and into the execution phase.

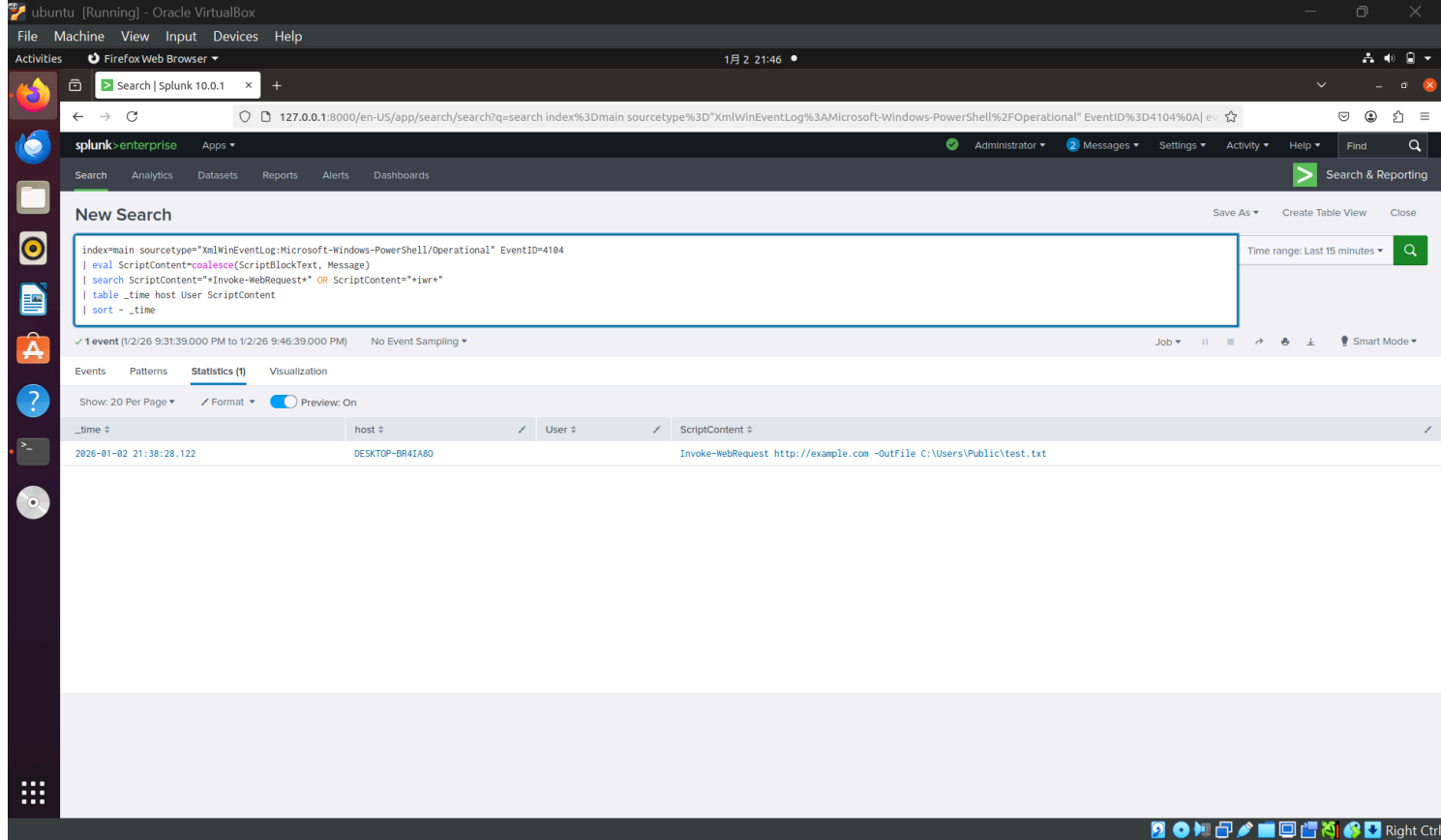


Figure 3: PowerShell Event ID 4104 showing Invoke-WebRequest activity used to download a file to the endpoint during the investigation.

Correlation Summary

After reviewing authentication and PowerShell telemetry, the individual findings were correlated to determine whether they represented isolated events or a single attack sequence.

The timeline revealed a sequence of events consistent with a potential attack chain.

The investigation began with multiple failed authentication attempts (Event ID 4625), followed by successful authentication events (Event ID 4624) on the same endpoint.

Shortly afterward, PowerShell Event ID 4104 logs revealed the use of Invoke-WebRequest to download a file to the system.

When viewed together, the evidence suggests a multi-stage attack sequence involving credential access, successful authentication, and post-compromise execution activity.

This correlation increased confidence that the events were related and formed part of a single incident rather than independent actions.

Investigation Timeline

Time (local)	Stage	Log source	Event ID	What happened	Evidence (screenshot/SPL)
21:13	1	Windows Security	4625	Multiple failed logons	4625 screenshot + SPL
21:28	2	Windows Security	4624	Successful logon occurs	4624 screenshot + SPL
21:38	3	PowerShell Operational	4104	Invoke-WebRequest download	4104 screenshot + SPL

MITRE ATT&CK Mapping

Stage	Technique	MITRE ID	Evidence
1	Brute Force / Password Guessing	T1110	Windows Security 4625 spike
2	Valid Accounts (logon)	T1078	Windows Security 4624 success after failures
3	PowerShell	T1059.001	PowerShell 4104 script block
3	Ingress Tool Transfer	T1105	Invoke-WebRequest download command

Investigation Limitations

During the investigation, several limitations were identified.

Sysmon Event ID 3 (Network Connection) telemetry was not available in the current lab configuration.

As a result, outbound network activity could not be directly validated through endpoint network telemetry.

Instead, evidence of external communication was inferred from PowerShell Event ID 4104 logs showing Invoke-WebRequest activity.

Future iterations of this investigation could be enhanced through:

- Sysmon Event ID 3 monitoring
- IDS visibility (Suricata)
- Zeek network telemetry
- Firewall log analysis

These additional data sources would provide stronger validation of outbound communications and attack progression.

Conclusion

This investigation demonstrated how authentication logs and PowerShell telemetry can be correlated to reconstruct a potential attack chain. The evidence showed a progression from failed authentication attempts to successful login activity and PowerShell-based file download behavior. This project reinforced practical SOC skills including log correlation, timeline reconstruction, MITRE ATT&CK mapping, and evidence-based incident investigation.